Jen-Li Kao

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Medical Imaging Systems | Computer Vision & ML | Embedded Signal Processing

EDUCATION

TEXAS A&M UNIVERSITY (TAMU) — **College Station, TX**
MS in Electrical and Computer Engineering — **Sept 2023 – May 2025**
- Awards: Graduate Merit Scholarship in Dept of ECE.
- Relevant Coursework: Digital Image Processing & Computer Vision, Pattern Recognition, Data Mining and Analysis, Analysis of Algorithms, MR Engineering, Advanced US Imaging Techniques, Bioelectromagnetism, Algorithm in Structural Bioinformatics

CHANG GUNG UNIVERSITY (CGU) — **Taoyuan, Taiwan**
MS in Computer Science and Information — **Jul 2022**
- Thesis: Breast Tumor Detection Using Multi-Channel Millimeter-wave 3D Imaging Technology: A Preliminary Study

NATIONAL DONG HWA UNIVERSITY (NDHU) — **Hualien, Taiwan**
BS in Computer Science and Information — **Jun 2018**

PUBLICATION

Conference Paper:
J. -L. Kao and Y. -P. Chao, *"Breast Tumor Detection Using Multi-Channel 62-69 GHz Millimeter-wave 3D Imaging Technology,"* 2022 IEEE 4th Eurasia Conference on Biomedical Engineering, Healthcare and Sustainability (ECBIOS), 2022, pp. 16-19, DOI: 10.1109/ECBIOS54627.2022.9945009

CONFERENCE

The 16th Symposium of Medical Imaging and Radiological Sciences (SMIRS)., Taoyuan, Taiwan, 22 Oct 2022
- Presentation, ***"Breast Tumor Detection Using Multi-Channel Millimeter-wave 3D Imaging Technology: A Preliminary Study."*** **- Honorable Mention Award**

RESEARCH EXPERIENCE

APPLIED MACHINE LEARNING RESEARCH, TAMU — **College Station, TX**
Research Co-op / Soumyajyoti Dutta (PhD Candidate, TAMU CSE) — **May 2025 – Present**
Project: Automatic Sigma Rule Generation Using Language and Vision Models
- Developed an end-to-end framework for automated Sigma rule generation by integrating **LLM reasoning**, **AST-based structural parsing**, and **multimodal security inputs**.
- Designed a **language-to-AST parsing pipeline** to evaluate rule consistency, structural complexity (Halstead and Cyclomatic), and syntactic correctness across varied model architectures.
- Prototyping a system that interprets heterogeneous security artifacts—such as log excerpts and visual indicators—and converts them into **machine-interpretable Sigma rul**es.
- Advancing to model training and performance evaluation, benchmarking natural-language and AST-guided models on the constructed dataset.

MEDICAL IMAGING PROCESSING AND COMPUTING VISION LAB, CGU — **Taoyuan, Taiwan**
Research Assistant /Advisor: Prof. Yi-Ping Chao — **Sept 2020 – May 2022**
Project: Breast Tumor Detection Using Multi-Channel Millimeter-wave 3D Imaging Technology
- Developed a breast tumor detection system using **millimeter-wave imaging** and **computer vision techniques** to **enhance high-resolution, non-invasive cancer imaging**.
- Designed **synthetic 3D phantoms** for breast, tumor, and skin-layer phantoms based on referenced studies to simulate real-world imaging conditions.
- Conducted **6 experiments** to evaluate system performance, including detection depth, minimum tumor size, and multi-angle imaging accuracy to **enhance 3D reconstruction quality**.
- Implemented **noise-reduction** and **beamforming methods**, including averaging and Delay-and-Sum (DAS), to reconstruct enhanced 2D/3D images and **improve overall spatial resolution**.

WORK EXPERIENCE

MEDICAL IMAGING PROCESSING AND COMPUTING VISION LAB, CGU **Taoyuan, Taiwan**
Research Assistant /Advisor: Prof. Yi-Ping Chao **Jan 2021 – Dec 2021**
Project: Mobile Health Technology Medical Device Pre-Market Review Trend Analysis

- Collected and analyzed pre-market guidelines for mobile medical devices and digital health regulations.
- Assisted **ASUS** and **Quanta** with pre-market compliance documents and compiled review Q&A.

CYLTEK LTD., **Hsinchu, Taiwan**
Intern **Sept 2020 – Feb 2021**
Project: Smart Healthcare Application Research Project

- Developed **Python-based real-time signal processing algorithms** using **millimeter-wave radar** for **contactless measurement of chest displacement, heart rate, and respiration.**
- Ported **Python-based DSP algorithms** to C++ for embedded integration and worked with hardware engineers to validate data-acquisition timing, sampling behavior, and system synchronization.
- Collaborated with **MediaTek and Chang Gung Hospital** to prototype healthcare devices.

TEACHING EXPERIENCE

COURSE: DATA STRUCTURE AND ALGORITHM, DEPT CSIE, CGU **Taoyuan, Taiwan**
Teaching Assistant /Instructor: Prof. Yi-Ping Chao **Fall 2021 – Spring 2022**

- Assisted in preparing lecture materials, grading, and counseling students on programming issues.

PROJECTS

Benchmarking Transformer-Based Metagenomic Functional Profiling ***Spring 2025***

- Compared **alignment-based** vs. **ML methods** for enzyme function prediction.
- Accelerated transformer inference with **CUDA** and benchmarked GPU performance on **HPRC** clusters.
- Analyzed **encoder substitution** and accuracy–balance trade-offs using **3-mer Jaccard similarity**.

Breast Cancer Imaging Classification: A Comparative Study of SVN, CNN, and Transformer Models on the CBIS-DDSM Dataset ***Fall 2024***

- Conducted a comparative study on **breast cancer classification** using the **mammogram dataset**.
- Built **SVM models with GLCM feature**, compared to **CNN and ResNet-based Transformers.**
- Applied **image preprocessing (normalization, contrast, augmentation)** to improve accuracy.

Integrated MRI Hardware and Pulse-Sequence System Using Analog Discovery 2 ***Fall 2024***

- Built a mini-MRI system using AD2 with **RF control**, **gradient encoding**, and **Python acquisition**.
- Calibrated RF and gradient hardware, performed shimming, and acquired optimized spin echoes.
- Developed 2D imaging using **PR** with **gradient encoding**, **k-space sampling**, and **FFT**.

Fashion MINST Modeling Comparison ***Fall 2023***

- Implemented and compared **CNN** and **ML models (LightGBM)** for imaging classification.
- Develop a robust **image preprocessing pipeline**, incorporating normalization, augmentation, and dimensionality reduction to optimize data for modeling.

TECHNICAL SKILLS

Core Competencies:

- **Imaging & Sensing Systems:** pulse/sequencing design, RF control, gradient/signal encoding, echo/signal acquisition pipelines, radar-based sensing
- **Computer Vision & Reconstruction:** multi-dimensional reconstruction, transform-based methods, projection techniques, and data preprocessing
- **Embedded DSP:** adaptive filtering, beamforming, signal synchronization, real-time DSP
- **Machine Learning:** CNNs, Transformers, model optimization, dataset curation, evaluation metrics
- **HPC:** CUDA acceleration, GPU profiling, parallelization, memory optimization

Technical Stack:
Python (NumPy, SciPy, OpenCV, scikit-learn, PyTorch, HuggingFace), C/C++, MATLAB, CUDA, TensorFlow, PyWaveforms/ AD2 SDK, Git, Linux, SLURM/HPRC